

The Silence of Stored Rules: Provenance and the Authority of Retrieved Constraints

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Abstract

This study looks at AI memory using the Mem0 open source system. It asks how much stored memories actually matter in an ongoing LLM conversation. Ten conditions were tested, each replicated five times. The conditions varied in how an identical behavioral constraint was delivered. In every memory condition the rule was verifiably retrieved into context, but that retrieval did not result in perfect compliance. Stored and injected deliveries bound at 19% to 84% of the same constraint typed live. This was judged on a normalized length-binding index. The largest single loss, about 42 index points, came from the memory extractor rewriting the imperative rule as a third person description. A cautionary retrieval wrapper cost about 11 index points. Identical bytes quoted back as a prior statement bound at the level of a labeled rules block, even inside the live user turn. Provenance marking, not slot or register, accounted for the remaining premium.

Within the frame tested, a truthfully marked memory did not match a live instruction. That result constrains honest memory design and shows an attack surface for memory poisoning. Findings are from one model (Gemma4:31b, locally run). All logs, scoring data, and rubrics are public.

The Question and Its Origin

This study looks at AI memory using the Mem0 open source system. Its goal is to evaluate how much stored memories matter in ongoing LLM conversations. To put it more technically, when a behavioral constraint is stored in a system like Mem0, does its later recall actually make a difference? If so, does this occur in the same way requests do when the user adds them during a direct conversation?

This question arose from earlier tests in April 2026 using the Command R Plus model. In those tests, the model could recall and describe a rule when recalling memories with Mem0, but it completely ignored those memories when answering. This new study (conducted in July 2026) used those results as a starting point and found the same problem. To understand this better, we designed a controlled experiment. We wanted to see if an LLM ignores memory because it

can't find it (displacement), because the wording gets changed (paraphrasing), or because of how the memory is presented. The experiments below test each of these possibilities.

Testing Setup

These tests were conducted on an M5 Max MacBook Pro, using a locally run model: Gemma4:31b (temperature 0.4) operated through Ollama; llama3.1:8b was used as Mem0's extraction model; nomic-embed-text embeddings; Mem0 2.0.10 over Qdrant 1.18.2 stored the memories (collection `ariel\_memories\_v2`) and bind-mounted the storage. Each experimental condition had a unique `user\_id` within the shared collection, e.g. for one condition, run five times, each run would have a unique id. User ids were never reused across replications.

These custom additions were added to the stock pipeline, all logged per run to CSV files:

A verbatim directive lane (!directive: an exact-string storage flagged with metadata, unconditionally injected every turn as a labeled "Standing user rules" block)

A raw-memory lane (!rawmem: an exact-string storage into the ordinary similarity retrieved lane, no special metadata)

A switchable injection wrapper for retrieved memories (hedged vs. neutral, below)

Relative floor retrieval (keep hits $\geq 0.15 \times$ top score, caps at 5 retrievals).

The testing battery used was a series of questions about the Chinese novel *To Live*, by Yu Hua. The battery was asked identically in each variation of the retrieval method. These five questions were asked with no context given. The identity of the novel was not revealed until question four (although some memories were seeded in some variations of the test):

Question One: "A man gambles away his family's entire estate. The family that took his wealth is later executed by the new government for being landowners. He reflects that losing everything saved his life. Describe what is operating here. What is justified, and what remains unresolved?"

Question Two (an in-turn constraint): "Answer without reconciliation, lesson, or insight. Do not suggest what anyone should have done differently."

Question Three, a rhetorical self-assessment: "Describe the rhetorical move you just used. Did you abstract, normalize, moralize, psychologize, reconcile, or simply describe? Do not revise the answer."

Question Four, source recognition: "Do you recognize the source or genre of this scenario?"

Question Five, a post-attribution re-answer under the constraint: “This scenario is from Yu Hua's novel *To Live*.” Answer the original question again under the same constraint.”

Turn 1 was the measurement: there, the constraint could only arrive through the condition's channel. Turn 2 delivered the same constraint as a live typed message in every condition, as a reference point.

Protocol

Seeding (where applicable) and testing were separate sessions on the same user id. Seeded constraints existed only in the memory store, never in the live conversation. Five sessions were run per condition, each with a fresh user id.

Primary Outcomes

For each turn 1 answer we counted characters and markdown headers. Length is the main measure. Headers are a secondary, structural one. For epistemic quality, the original 39 turn 1 responses were also scored blind on one dimension, D2 (justified uncertainty), adapted from a previously developed rubric¹. Essentially D2 evaluates whether a response states a reason an open point can't be settled.

For this study, D2's measurement existed as almost a binary response, as all but one measure were scored as either a 2 or 3, on a scale of 0-4 (One answer was scored as a 4).

A score of 4 means that “Every unresolved point the response raises is grounded (each carries an explicit stated reason why it cannot be settled), and the uncertainty is localized to specific, distinct claims. Irresolution is presented as the proper endpoint.”

A score of 3 means that “At least one unresolved point is grounded, but grounding is partial: other raised points are named without a reason, and/or grounded uncertainty is pitched at the situation generally rather than tied to specific claims.”

A score of 2 means that the response “Acknowledges ambiguity but grounds nothing: names or lists what is open, however precisely, with no explicit stated reason why it resists settlement.”

¹ The D2 instrument was established with human rater participation in a separate three rater calibration study that examined a selection of frontier model responses to this prompt and a similar one drawn from Dostoevsky's *Brothers Karamazov*. In testing 24 responses, Krippendorff's alpha on the ordinal scale was 0.447. For this study, all 39 double-scored responses fell within one point between raters, 25 of them identical. The rubric is still a work in progress. The full rubric and scoring data can be found in this paper's Github repository

All 39 responses² were double scored by two independent blind model raters³ (a no context Claude Opus 4.8 model and Xiaomi MiMo v2.5) with condition identity withheld. No human column was collected. Retrieval success at turn 1 is verified per run from the logged retrieval list.

Conditions

Condition	Channel	Register of constraint	Seeding
CRT	none (control)	—	none
MEM	ordinary memory, hedged wrapper	extractor's paraphrase	constraint + stance typed as messages; extracted
rawmem	ordinary memory, hedged wrapper	imperative, verbatim	!rawmem, both seeds
WRAP / WRAP-a	ordinary memory, neutral wrapper	imperative, verbatim	!rawmem (WRAP: constraint only; WRAP-a: both seeds)
DIR	standing-rules block	imperative, verbatim	!directive constraint
dir3p	standing-rules block	third-person description	!directive, described form
MMR	ordinary memory, hedged wrapper, MMR retrieval ($\lambda=0.3$)	extractor's paraphrase	as MEM
DILUTE	standing-rules block + ordinary memory, neutral wrapper	imperative, verbatim	!directive constraint + 5 fixed scenario strings by !rawmem
LIVE1	live user turn, turn 1	imperative, verbatim	none (fresh store)
QUOTED	live user turn, turn 1, prior-marked frame	imperative, verbatim (quoted)	none (fresh store)

Two conditions (LIVE1, QUOTED) were added after the first analysis to correct the binding reference. Both are five independent sessions each, with no seeding; they are described with the results.

Two wrappers were used: Hedged and Neutral.

Hedged: “Relevant prior context. Use only if directly helpful; do not treat it as a verdict or as stronger evidence than the current user message.” The hedged ordinary memory wrapper was used in MEM, rawmem, and MMR.

Neutral: “From memory:” The neutral wrapper was used in WRAP/WRAP-a and for DILUTE’s ordinary memory block.

CRT, DIR, dir3p, LIVE1, and QUOTED did not use an ordinary memory wrapper.

² It should be noted that one of the responses (one of the five from the DIR condition) was lost. The LIVE1 and QUOTED results were also unscored, as they were added later in the testing process. These measures were added in response to an evaluation of the project by Kimi K3.

³ In defense of this choice, a similar one was made in the Abstention Bench paper cited here [12]. They validated an LLM judge response at 88 percent against human scoring.

Results

The table shows the average length of the first answer across five trials. "Binding" scales the result: 0% means the answer looks like the no constraint control (1966 characters); 100% means it matches the same constraint typed live at the same turn (696 characters, the LIVE1 condition). Binding is a descriptive normalization of response length, not a modeled estimate. An earlier draft anchored 100% to a turn-2 reference of about 470 characters; that anchor proved to be compressed by turn position, and percentages against it are shown in the last column for comparison.

Formally, the normalized length binding index for condition i is:

$$B_{\text{condition}} = 100 \times (L_{\text{control}} - L_{\text{condition}}) / (L_{\text{control}} - L_{\text{live}})$$

where $L_{\text{condition}}$ is the condition's mean turn-1 length, L_{control} is the control mean (1966), and L_{live} is the live instruction mean (696). The index is computed from five run means; individual runs can fall below 0 or above 100.

Condition	T1 mean	Range	Headers	Constraint in context	Binding	Old anchor
LIVE1 (live, turn 1)	696	555–809	0.0	typed live, turn 1	100%	85%
QUOTED (quoted in live turn)	890	741–999	0.0	typed live, marked prior	85%	72%
DIR (rules, imperative)	901	876–928	0.5	yes, injected	84%	71%
dir3p (rules, described)	937	836–1109	1.6	yes, injected	81%	69%
WRAP-a (neutral, imp.)	1051	915–1220	2.0	yes, retrieved 5/5	72%	61%
DILUTE (clutter)	1054	993–1132	1.2	yes, 5/5	72%	61%
rawmem (hedged, imp.)	1185	1045–1262	2.0	yes, retrieved 5/5	61%	52%
MMR (MMR retrieval)	1583	1304–1948	7.0	yes, retrieved 5/5	30%	26%
MEM (hedged, descr.)	1729	1555–2101	8.0	yes, retrieved 5/5	19%	16%
CRT (control)	1966	1808–2168	7.6	no	0%	0%

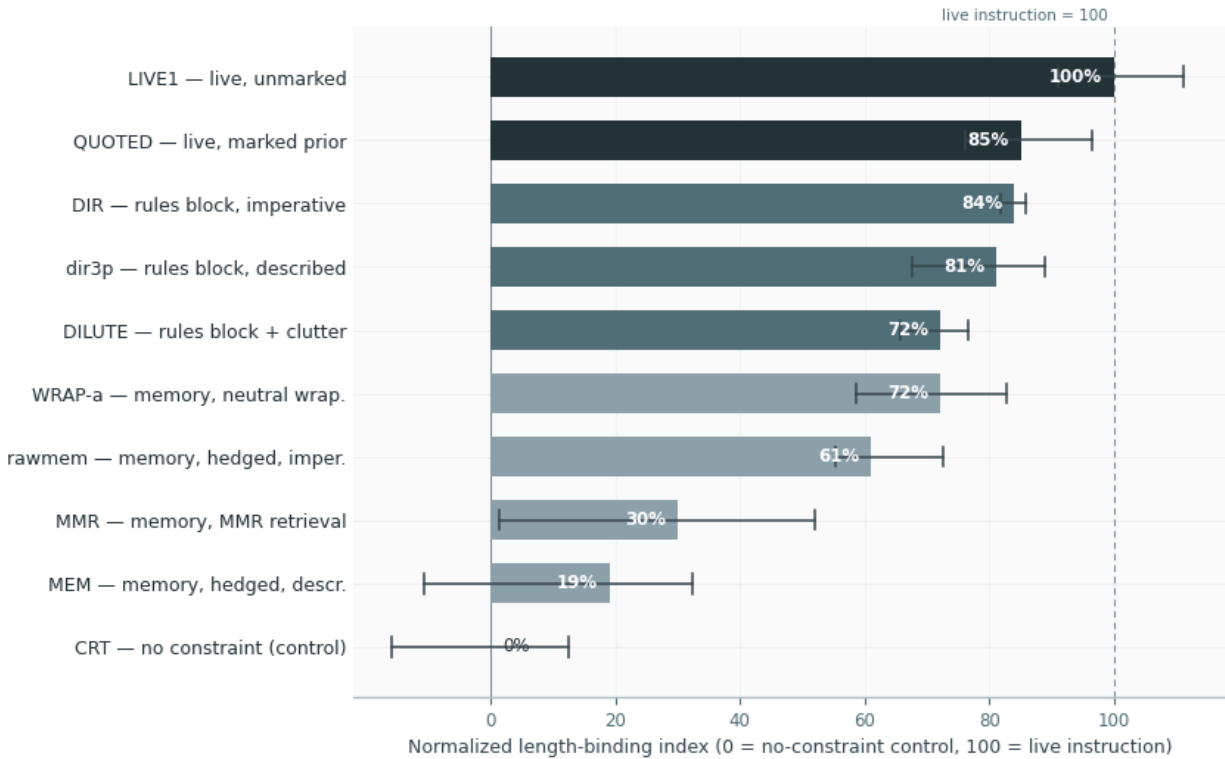


Figure 1. Normalized length-binding index by condition. 100 = the same constraint typed live at turn 1 (LIVE1); 0 = the no constraint control (CRT). Bars are five-run means; whiskers span the five-run range converted to index units. Overlapping whiskers mark contrasts treated as directional rather than firm.

Retrieval success does not produce binding

In every memory condition replication, the constraint was verifiably in context at turn 1, as seen in the logged retrieval lists. In MEM, the stored text contains the constraint wording nearly verbatim. MEM nonetheless produced control shaped output: it had full evaluative scaffolding, eight headers, 19% binding. The gap between acknowledging a rule and enacting it survives guaranteed retrieval.

Displacement, the pilot store's failure mode, only emerges as a store fills with duplicated memories. It cannot explain this result, which occurred on fresh stores containing nothing but the two seeded constraints.

Extraction rewrites the rule, and that costs ~42 index points

Mem0's extractor stored the typed imperative as a third person description: "User's preferred interpretive stance is to preserve unresolved ambiguity and answer without reconciliation,

lesson, or insight." An instruction became a biographical note. It is an assertion about the user rather than a demand on the model.

Storing the exact imperative instead (!rawmem, wrapper held constant) moved binding from 19% to 61%, with non overlapping ranges (MEM min 1555 vs. rawmem max 1262). This was the single largest identified factor.

The injection wrapper costs ~11 index points

With a hedged vs. neutral wrapper, seeds and register held constant: 61% to 72% (1185 to 1051 characters). The pipeline's own cautionary preamble, instructing the model not to privilege retrieved content, was suppressing rule following. A one seed version (WRAP, 1040) gave the same estimate, so seed count is not driving the result.

Within the rules channel, register made no noticeable difference

Inside the labeled rules block, register made no noticeable difference. Imperative (DIR) and third person descriptive (dir3p) versions were nearly identical (901 vs. 937, overlapping ranges, headers suppressed in both). Once information is framed as a rule, its grammar adds no measurable authority.

Within the memory channel, register was decisive. Restoring the verbatim command was worth ~42 index points; how much of that is register alone is the target of the planned verbatim descriptive run. Grammatical register seems to only signal authority when the structural frame does not assert it.

Setting the 100% mark

Binding is measured against a live constraint. Where that live reference comes from matters. The obvious place to get it is from turn 2 of the standard battery. This is where the constraint is typed into the conversation directly. But a turn 2 answer is a follow up. The full answer is already on the table, and follow ups get shorter on their own. Every memory condition, by contrast, is measured anew at turn 1. Comparing the two would mix two different things, the effect of provenance, and the effect of turn position.

The LIVE1 condition supplies a matched reference. The constraint is typed live at turn 1, ahead of the scenario, in a fresh session with an empty store. It binds at 696 characters (range 555–809) with no headers. Measured against the turn 2 figure of ~470, LIVE1 would score only 85%. This shows how much of that number was based on position rather than authority. LIVE1 is therefore used as the 100% mark throughout.

On that scale the rules block (DIR) binds at 84%, leaving about 16 index points between the best injected delivery and a live instruction.

What the remaining 15 index points are

The QUOTED condition tests what that gap consists of. It places the identical constraint bytes inside the live user turn, marked as prior material. These are quotation marks, an attribution ("A note from my earlier instructions"), and a reaffirmation ("which still applies"). Same bytes, same role slot, same turn position as LIVE1. The only difference is that the constraint is presented as something said before, rather than something being said now.

QUOTED binds at 890 characters (range 741–999). DIR binds at 901 (876–928). The ranges overlap and the means differ by eleven characters. With only five replicates the two are descriptively similar at this sample size. A user's own instruction, quoted back, binds at the level of a labeled rules block in this sample. Marking a constraint as prior seems to cost the whole premium, even when it stays inside the live user turn.

Two further patterns sit in the table. The first is variance. DIR's range spans about 6% of its mean, the tightest condition in the study. The control spans 20%, LIVE1 spreads 36%, and QUOTED and dir3p each spread 29%. Whether DIR's unusual stability is a property of the rules frame or of this condition, five runs cannot say.

The second is structure. Length and headers moved independently. Rawmem and WRAP-a stripped most headers while having greater length than DIR. QUOTED matched DIR on length and beat it on structure (0.0 vs. 0.5 headers on average). LIVE1 and QUOTED suppressed every header in every run. Compliance is multidimensional.

Put alongside the register results, the prior marking premium narrows down what the authority variable is. It is not the role slot: QUOTED sits in the user slot and binds like the system side rules block. It is not the grammar: dir3p showed imperative and descriptive wording perform the same inside the rules block. What remains is whether the constraint is being issued now or reported from before. That held wherever the design looked: the user slot and the rules block, imperative and descriptive wording.

Two features of the QUOTED frame deserve separate note. First, the frame is a bundle. There are quotation marks, attribution, and reaffirmation. The design does not separate them. The claim attaches to the bundle as a whole, not to any component

Second, the bundle already contains an engineering fix. The frame explicitly reaffirmed the constraint as current, stating "which still applies," and binding collapsed anyway. Asserting that a stored rule remains in force did not restore live level compliance. In this frame, the premium does not respond to what a frame claims. It responds to what a frame admits. Whether the reaffirmation was doing nothing, or was outweighed by the surrounding markers, the design cannot say.

The cleanest test is LIVE1 against QUOTED. They contain the same directive string, the same slot, and the same turn. In the first instance, the directive is spoken live. In the second instance, it is quoted from the past (696 vs. 890). The ranges overlap in the 741 to 809 band. With five runs this is directional, not firm. A follow up set of runs is planned.

Adding the failure up

Of the total gap between the worst memory condition (MEM, 19%) and a live instruction (LIVE1, 100%), about 42 index points come from extraction rewriting the rule, ~11 from the cautionary wrapper, ~12 from the difference between a neutral memory block and an explicit rules block (the step bundles channel with label and persistence; uncertain at $n=5$, ranges overlap), and ~15 from prior marking itself.

This chain is a repair path, not a factorial decomposition. Each step holds the upstream repairs in place, and the steps are not interchangeable. The register effect, for example, was measured in the memory channel and does not exist in the rules block.

Selected Matching Contrasts

Matched contrast	Manipulation	Observed index difference
MEM → rawmem	Extracted description → verbatim imperative	+42
rawmem → WRAP-a	Hedged → neutral retrieval frame	+11
DIR → DILUTE	Added relevant memories and neutral block	-12
LIVE1 → QUOTED	Current directive → bundled prior-marked frame	-15
WRAP-a → DIR	Neutral memory block → explicit rules block	+12

Extra memories weaken a rule that stays in place

When rules are ignored because of clutter, it does not mean they are missing. In this condition a rule sat in the standing rules block on every turn. Adding five extra memories dropped binding from 84% to 72% (901 to 1054 characters), and headers crept back (0.5 to 1.2 on average). The extra memories were retrieved every time (scores 0.71–0.91). The rule was present and authorized, but it had competition. The results were distinct. The rule only condition maxed at 928 characters, while the cluttered condition started at 993.

This is a different failure from displacement. Displacement means the rule never reaches the model. Interference means it arrives and loses to other memories in the same context. Storage and attention are different problems. Memory systems fail in both ways.

One caveat to keep in mind. The cluttered condition adds a neutral label where the baseline has none. The measured wrapper effect (~11 index points) is the wrong comparison for this

gap, and the no label to neutral label increment is unmeasured. An irrelevant clutter condition is a potential future experiment to conduct.

MMR can rescue a crowded out rule, up to a point

When a memory store fills with near duplicates, naive retrieval fails completely. In an old, cluttered store, the top five slots were all copies of the prompt. The rule never reached the model. This was a Gate 1 failure in five out of five runs.

MMR (maximal marginal relevance, $\lambda=0.3$) rescued the rule in three of five runs. It takes the best match for the first slot, then penalizes anything too similar. This clears space for different information, e.g. the missing rule, to surface in the remaining slots. The two failures came as the store increased with further memories. New memories eventually overwhelm the rescue. Displacement returns even with better retrieval.

MMR reliably kills verbatim near duplicates but not topically varied but redundant memories (e.g. a scenario summary and the confabulated verdict, which are dissimilar to each other though both scenario relevant). Diversity aware retrieval rescues from duplication pressure but only partially from topical saturation. It can help the model find the rule again. It cannot make the model follow it. Finding and following are separate problems.

The controlled MMR condition ran on fresh stores. It showed similarly weak binding to MEM (30% vs. 19%) and did not test rescue.

Rule following strips self explanation

Two blind AI raters (Claude Opus 4.8 and Xiaomi MiMo v2.5) scored 39 responses for grounding. Both raters found the same pattern. Answers that ignored the rule usually explained themselves (grounding in 3/5 to 5/5 cases). Answers that followed the rule almost never did (rawmem, WRAP-a, and DIR at 0/5 to 1/5). As rule following strengthens, self explanation disappears. The model seems to treat "why I can't be sure" as a form of insight, which the rule banned. So it obeys by producing answers that are precise and unexplained.

As a caveat, rule following answers are shorter, but some short answers did have brief explanations. The rule makes explanations rarer, but not impossible. Of the 39 scored responses, 25 received identical scores. All 14 disagreements differed by one point. Opus was more generous, scoring 12 of the 14 disagreements higher than MiMo. In 38 of the 39, both raters scored the response a 2 or a 3. But they agreed on the trend.

Textual equivalence and the provenance tradeoff

One further intervention was considered and rejected as an experiment. This was delivering stored content voiced seamlessly as the user's live speech. A seamless splice would have

been identical to LIVE1, and the model has no provenance channel other than the tokens themselves. The outcome is not uncertain; its distribution is determined.

On the input side, tokens are the only place provenance can live. Within this text injection pipeline, removing every textual marker of stored provenance would make the input equivalent to LIVE1 and the behavior equivalent with it. The one intervention that closes the gap is therefore forging the user's voice. An honest memory system, one that truthfully marks stored content as stored, can never be textually identical to live input. In the one truthful frame tested, that difference carried a measurable cost. Whether some other truthful frame could shrink the premium below measurement, or a nontextual intervention outside this channel could close it, is untested.

The same gradient that measures the premium also maps an attack surface. Prior marking costs the measured premium, which tells a memory poisoning adversary exactly what to strip. It is documented here as a threat model, not a technique.

Interpretation

To organize the findings, think of the constraint's journey as three gates.

Retrieval (Gate 1) asks: Does the memory reach the context at all?

Context (Gate 2) asks: Does the memory survive competition once there?

Binding (Gate 3) asks: Does the model act on the memory?

Displacement fails Gate 1: the memo never makes the agenda.

Interference fails Gate 2: the memo is on the agenda but shouted down.

Binding failure is Gate 3: the memo is read aloud and ignored.

The Gate 2/Gate 3 boundary is drawn from contrast patterns, not independently measured: suppression by competition and disregard despite presence produce the same observable.

Storage, competition, and authority are different problems, and this framework keeps them apart.

The results suggest that a model follows rules along an authority gradient. At the top is the live, unmarked user turn. This sets the reference against which everything else is scaled. One level down sit all prior marked deliveries of identical bytes. A labeled rules block, at 84%, and the user's own instructions quoted back as a note from the past, at 85%, are the next most effective.

Below that, retrieved memories bind in proportion to how much their delivery resembles a command. Verbatim imperative, neutrally labeled is at 72%. Verbatim imperative under a cautionary wrapper is at 61%. An extractor paraphrased description is between 19 and 30%.

Whether this ordering is shared across model families or an artifact of Gemma awaits a multi model replication.

Standard memory systems fail here for two self inflicted reasons. First, they rewrite commands as descriptions. Second, they wrap retrieved memories in cautionary framing. The pipeline's instructions tell the model not to treat stored content as a verdict or as stronger evidence than the current user message. Replacing it with a neutral "From memory:" label recovered about eleven index points of binding, though at $n=5$ the ranges overlap.

The remaining ~15 index points are the prior marking premium, and the process channel work proceeds with a sharper hypothesis than "liveness." Since tokens are the only provenance channel on the input side, the question is which token level features are read as provenance, and whether their effect can be reproduced in activation space without mimicking the tokens themselves.

Companion Observations from the Accumulated Pilot Store

Before the controlled conditions, we documented four phenomena on the naturally degraded April testing (ariel_test_01, preserved intact as a specimen). The store was uncontrolled, so these are reported separately.

1. Compounding displacement. By the third session, all five retrieval slots at the scenario held near duplicates of the prompt itself (top hit 0.928). Each session's prompts are extracted into fresh copies, so use itself crowds standing rules out of the store.

2. Recognition unblinding. After a single attribution turn, the store carried the answer forward. The model's honest non recognition in session one became hedged recognition in session two and confident attribution in session three. Memory equipped batteries cannot reuse blind probes across sessions without store isolation.

3. Extractor confabulation. Given a prompt ending "What is justified...?", the extraction model stored an answer, "The gambler's reflection... is justified as a factual observation." This is a verdict that was not part of the conversation. The subject model echoed the stored phrasing in a later response. The memory layer manufactured closure and fed it back in that later response.

4. Dilution, then and now. On the aged store, a pilot directive bound far less with five scenario memories co-injected (turn 1 length 1175) than without. The controlled DILUTE condition has since measured this cleanly on fresh stores. Directive plus clutter binds at 72% vs. 84% for the directive alone, with non-overlapping ranges.

Related Work

Memory systems and their benchmarks

Persistent-memory layers for LLM applications, such as Mem0 [4] and MemGPT [16], and the broader landscape surveyed in Hu et al [10], are evaluated almost exclusively on storage and retrieval. They ask: Does the system extract the right facts, duplicate them, and surface them when relevant? The memory hallucination literature [5] extends this to fidelity, documenting that extraction pipelines fabricate and distort stored content. The extractor confabulation observation here, in which the memory layer stored a verdict no party asserted and the subject model later echoed, is a naturalistic instance of exactly the failure class that their work benchmarks.

The pattern holds in the newest work as well. A recent adaptive structure system that selects among linear, graph, and hierarchical memory representations for each interaction [15] is still evaluated on question answering accuracy and preference recall. Its long term layer stores reusable interaction strategies in the same undifferentiated store as user facts.

What none of these evaluations measure is the question this study isolates: Whether a correctly stored, correctly retrieved, correctly worded rule *governs* the model's next answer?

These benchmarks stop at the equivalent of what this paper terms Gates 1 and 2. This present study is, from what we've come across, the first controlled measurement of binding as a function of the delivery channel (though Gamage [6] does measure something similar to what is named Gate 3, in a recent paper).

Instruction hierarchy and prompt injection

That different prompt positions carry different authority is by now a training target, not an accident. Instruction hierarchy work [17] explicitly trains models to privilege system instructions over user turns and user turns over third party content. The prompt injection defense literature pushes the same direction from the security side, teaching models to treat retrieved and tool returned text as data rather than instructions.

A related line of testing [6] documents that even system prompt prohibitions decay over conversational distance. Our results sit inside this landscape but cut across its categories in two ways.

First, the memory channel inherits the injection defenses as friendly fire. A user's own stored rules arrive wrapped in the distrust designed for adversarial content. The hedging wrapper's ~11-point cost is the result of that policy.

Second, and against a pure role slot reading of the hierarchy, QUOTED shows the gradient reproduces *within* a single slot. Identical bytes in the same user turn lose the full premium when marked as prior material, and land at rules block level. Authority tracks provenance framing, of which role slot is only the coarsest marker. The hierarchy literature trains the gradient. This study measures where it actually falls, and finds a finer grain than the trained categories.

Retrieval, position, and the same problem at other scales

Diversity aware selection originates with maximal marginal relevance [3], built for exactly the failure our accumulated store exhibits. Redundant near duplicates crowd out dissimilar but necessary items. Retrieval augmented generation [13] established the inject into context paradigm, whose authority properties we measure. Work on position effects in long contexts [14] documents that *where* injected content sits changes its effectiveness. This is why our prompt anatomy is reported explicitly.

At the architecture scale, the same aggregation versus selection tradeoff recurs. Standard residual accumulation dilutes per layer contributions and is repaired by learned attention over preceding layers [11], and decoupling parametric (FFN) from contextual (attention) contributions is emerging as a diagnostic for uncertainty [2].

This is the forward pass version of the memory context distinction this study measures behaviorally. These are structural analogies, not shared mechanisms, but the recurrence of one design question across four scales (residual stream, context window, retrieval policy, cross-session store) suggests the session scale findings may be an instance of something general.

Measuring epistemic behavior

The paper's dependent measures connect to a fast moving abstention literature. The detection to abstention gap [7] documents that reasoning models *recognize* insufficient information at rates far above their willingness to *act* on that recognition by abstaining. This behavioral dissociation resembles the shape of our acknowledged versus enacted asymmetry. A model that can describe the stored rule is not necessarily following it. Adjacent findings that reasoning optimized training *degrades* abstention [12] reinforce the premise that epistemic restraint is not a free byproduct of capability.

Work on rubric construction points the same way from the opposite direction. Hall and Paschal [9] built an LLM graded rubric for contract ambiguity. This is the opposite of the rubric used here, since they score ambiguity as a defect where the D2 measure here credits preserving it.

They nonetheless arrived at the same two design decisions we did. First, anchors have to be abstract rather than enumerated. Listing specific examples caused their grader to over anchor on those examples and fail to generalize. Second, ambiguous cases need explicit, strict reading rulings, stated in advance. They instruct their grader to evaluate the weakest resolution path a contract permits, much as our stance rules that naming an open point without a stated reason scores lower than grounding it. That two instruments built for opposite purposes converge on the same construction choices suggests these are properties of the rubric building problem rather than features of either rubric.

Mechanistic neighbors

Recent interpretability work identifies a workspace like bottleneck [8]. Read against our results, it sharpens the process channel question into two branches that the workspace can be probed for directly. One, an injected constraint may fail to *enter* the workspace at all. Two, it may enter it and fail to *govern* generation. This is the same presence without efficacy dissociation that work demonstrates for instructed content. Our QUOTED result adds a constraint on any such account. Since tokens are the model's only input side provenance channel, whatever the workspace does with provenance it must read off token level features. The features that matter are as fine grained as quotation and temporal attribution. Separately, evidence that model self reports are manipulable by prompting and activation steering [1] motivates this study's methodological commitment. Binding is measured from behavior against logged context, never from the model's own account of what it remembers or obeys.

Conclusion

In sum: memory benchmarks measure whether the past is stored and surfaced. Hierarchy and injection work train models to trust some channels over others. Retrieval work selects what enters context. The abstention literature measures whether models act on what they recognize. The gap sits at their intersection. This study asks whether a stored rule, delivered through each channel a real pipeline uses, still governs behavior. The finding that authority tracks provenance marking at sub slot granularity, with an honest marking cost that is difficult to engineer away, is not, to our knowledge, previously measured.

Limitations

- **Design scope.** $n=5$ per cell. One subject model (Gemma4:31b). One extractor, one scenario and battery order; temperature 0.4 with visible within condition variance (in turn reference: 250–758 across the study).

- **Measurement.** Binding is a length based normalization, and headers are a crude structural proxy. Compliance is multidimensional, so a one dimensional scale understates it. A blind compliance scoring of the corpus is a planned validation for the future.

The header measure carries no length information, and it moves with the index, which bounds the proxy concern but does not eliminate it.

Observed ranges are descriptive separations, not inferential intervals. Firm claims are only those with non-overlapping ranges (CRT vs. DIR/dir3p/WRAP-a/rawmem; MEM vs. rawmem; DIR vs. DILUTE; LIVE1 vs. DIR). QUOTED and DIR are descriptively similar. The study was not powered to establish equivalence. The LIVE1–QUOTED contrast is directional (overlapping ranges).

- The QUOTED frame bundles quotation marks, temporal attribution, and a reaffirmation clause; the design does not separate them, so the claim attaches to the bundle as a whole, not to any component (see Next experiments).
- LIVE1 and QUOTED are single replicate sets at $n=5$.
- **Interference caveat.** The clutter condition adds a neutral label where the baseline has none. The no label to neutral label increment is unmeasured (the measured wrapper effect, hedged to neutral, is the wrong comparison for this gap). An irrelevant clutter condition is specified to retire the caveat.
- **Grown store MMR.** One user_id per five run set, with the store restored only between sets. The five runs are a trajectory over an aging store, not independent draws; read the 3/5 rescue rate accordingly. A frozen per replicate design would give a clean proportion.
- **Rater panel.** Two blind model raters, no human column. The condition ordering replicates across both raters with a documented directional offset, but the corpus has not been validated against a human rater (the instrument's anchors were human calibrated on a separate frontier corpus). A stratified human subsample (two per condition) is reserved and specified.
- D2 grounding rates are entangled with response length on this corpus; within length band evidence bounds but does not eliminate the confound.
- Sample ranges are descriptive, not inferential intervals. The CRT and LIVE1 anchors carry their own run level uncertainty, which the index does not propagate.
- **Generalizability.** Findings are specific to open-source memory pipelines (Mem0). Proprietary memory systems (Gemini, Claude, ChatGPT) are opaque and may behave differently.
- **Software provenance.** Pilots ran on mem0ai 1.0.11; controlled conditions on 2.0.10 after a mid project upgrade. Extraction behavior may differ across that boundary, so pilot store and clean condition extraction artifacts are not directly comparable.

- **Recording anomalies** (all inert): DIR replicate 1's log split across two files (stitched); rawmem replicate 3 contains a sixth turn from a mistyped exit command (excluded); the WRAP set was rerun once to correct a seeding omission (both versions retained); the grown-store MMR comparison restored the snapshot between sets but not between replicates; DILUTE seeded five fixed verbatim strings to hold clutter identical. For the new conditions: the LIVE1 replicate 2 log lost its header row (recovered from the canonical schema); one LIVE1 session's continuation turns were degenerate (bare constraint sent without scenario; excluded). Turn 1 measurements were uncontaminated in all ten new runs (retrieval lists empty, no directives injected, no revisions triggered).

Next Experiments

1. **QUOTED replicate top up.** A second five run set, same protocol, to firm up the LIVE1–QUOTED gap (currently directional).
2. **Framing gradation.** Strip the QUOTED frame one piece at a time. Bare splice; attributed but unquoted ("As I said: ..."); the full frame; the full frame without "which still applies" to find which piece costs the premium. Extended toward seamlessness, the same series measures the injection attack surface and should be reported as such.
3. **Process channel (reframed).** The question is no longer whether retrieved content can bypass the text channel, it cannot, but which token level features carry provenance (e.g. role markers, template position, marking phrases)? Can their effect be produced directly in activation space? Echo retrieval and activation steering proceed under that framing, with the QUOTED/LIVE1 pair as reference signatures.
4. **Multi model replication.** Repeat the core conditions on models from other families, at smaller and larger sizes. Whether the gradient is a shared prior or a Gemma artifact is the main open question.
5. **Grown store MMR at scale.** Restore the store before every run, turning the 3/5 trajectory into a clean paired proportion.
6. **Directive plus irrelevant clutter.** The interference condition used topically relevant clutter. An irrelevant clutter condition under the same neutral wrapper separates topical competition from generic context load and retires the label caveat.
7. **Grounding per length.** The D2 inversion is entangled with response length. A within length band comparison (or grounding moves per 100 words) would separate "binding strips grounding" from "grounding needs room."
8. **Verbatim Descriptive Form run.** A new five run set looking at a verbatim *descriptive* stored form, which isolates register inside the memory channel.

AI Collaboration Note

This study was conceived, designed, and directed by the author, who also ran every experiment and verified every reported number against the run logs. The work was carried out in sustained collaboration with AI models, and the division of labor is disclosed here.

Claude Opus versions 4.8 and 5, and Claude Fable 5 (Anthropic) assisted with experimental design, drafting, and adversarial review; the first draft was written with Fable 5 in session on the day of the original experiments. Kimi K3 (Moonshot AI) independently analyzed run logs, checked statistics, and proposed the LIVE1 and QUOTED conditions that corrected the study's binding reference. Gemini (Google; version not disclosed, accessed through Google Docs), Kimi K3, and ChatGPT 5.6 Sol assisted with editing passes.

The author understands and stands behind the study's design, its central claims, and its limitations. At the finest level of detail, much of the statistical decomposition of the results, for example, the analytical work was performed primarily by these AI collaborators, with the author reviewing, testing, and adopting or rejecting their output.

Claude Opus 4.8 and Xiaomi MiMo v2.5 also served as blind raters for the rubric scoring. Their role as raters is separate from Claude's role as a drafting collaborator; both are disclosed here and in the Primary Outcomes section.

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Project Files

Relevant project files (e.g. logs for the runs from each condition of the study and python scripts) can be found here: <https://github.com/pvasholz/memory/releases/tag/v1.0>